Apiwich Sumeksri

(416) 930 3290 | apiwich.sumeksri@mail.utoronto.ca | linkedin.com/in/apiwichsumeksri | github.com/apiwichs | apiwichs.github.io

Education

University of Toronto
Sept. 2024 – Jun. 2028

BASc, Electrical and Computer Engineering — Minor in Artificial Intelligence

- 3x Dean's Honour List (Fall 2024, Winter 2024, Winter 2025)

Technical Skills

Languages: Python (NumPy, Pandas, Matplotlib), C, C++, Verilog, Assembly, Java
Tools & Hardware: Altium Designer, Git, Oscilloscope, Logic Analyzer

Experience

Analog Devices, Inc.
May 2026 – Present

Engineer Intern

- Responsible for improving the VI (Voltage-Current) Card to extend its operating voltage range from ±50V to ±100V while maintaining precision and low-noise performance.
- Setting accurate circuit biasing points to ensure stable and linear operation of analog components across the extended voltage range.

University of Toronto Thai Student Association
Jun. 2025 – Present

Events Associate

- Built and queried an SQL-based attendee database to group registrants by event, arrival time, and role, enabling faster check-in and smoother on-site coordination.
- Acted as point of contact between executives, volunteers, and attendees during events, resolving real-time issues and adjusting task assignments to maintain smooth operations.

Provectus Networks
Nov. 2022 – Mar. 2023

IT Engineering Intern

- Upgraded client and institutional workstations by replacing outdated DDR3 RAM and hardware components, **reducing boot and loading times by up to 35%**.
- Diagnosed and resolved technical issues including hardware failures, driver conflicts, and OS errors across 20+ devices.

Design Projects

Hexforge Arena

Digital Systems Designer

- Built a real-time top-down PvP fighting game in bare-metal C on the DE1-SoC with no OS or graphics libraries, directly interfacing with memory-mapped VGA for all rendering.
- Implemented a dynamic camera system tracking both players and adjusting zoom based on distance, using nearest-neighbor sampling for manual pixel scaling to the VGA buffer.
- Designed class-based combat (Warrior and Sorceress) with animation state machines, hit detection, and real-time input handling, all coordinated within a single bare-metal loop.

CPU Pipeline Performance Simulator

Systems Programmer

- Developed a cycle-accurate 5-stage CPU pipeline simulator modeling IF/ID/EX/MEM/WB execution, instruction retirement, and pipeline drain behavior.
- Implemented RAW data hazard detection and stall insertion logic to simulate pipeline bubbles and dependency-driven performance penalties.
- Analyzed CPU pipeline performance by computing CPI, stall rate, and stage occupancy from cycle-level execution data using Python (NumPy, Pandas).
- Generated per-cycle performance traces and visualized pipeline occupancy and stall behavior using Matplotlib to study execution efficiency.

Wayout 1984-Inspired FPGA Maze Game

Digital Systems Designer

- Delivered an FPGA-based first-person maze game over a 3-week development cycle, integrating clock dividers, pixel-timing generators, and VGA output.
- Developed control logic using shift registers, finite state machines (FSMs), and synchronous update cycles to manage player movement, wall detection, and state transitions.
- Verified system behavior through ModelSim waveform analysis, register-level debugging, and cycle-accurate simulation to ensure reliable frame timing, memory access, and collision-detection accuracy.

Soundproofing of Elevator-Adjacent Rooms

Project Manager

- Led a 5-member team on an acoustic noise-reduction design project, earning recognition as one of the top 28 projects out of 187 teams in the faculty showcase.
- Implemented structured Gantt-chart planning, improving workflow efficiency by 30% and enabling the team to deliver the final design 2 weeks ahead of schedule.
- Developed a requirements document translating stakeholder needs into functional specifications, performance targets, and engineering constraints, providing the foundation for all design decisions.